

Decent Coaching Classes

IX CBSE

Worksheet

Mathematics | Chapters 1–7

SECTION A — Multiple Choice Questions

Chapter 1 · Orienting Yourself: The Use of Coordinates

1. The point $(-3, 5)$ lies in the
(a) I quadrant (b) II quadrant (c) III quadrant (d) IV quadrant
2. The distance of the point $(6, -8)$ from the origin is
(a) 6 units (b) 8 units (c) 10 units (d) 14 units
3. The mid-point of the line segment joining $(2, -4)$ and $(8, 6)$ is
(a) $(10, 2)$ (b) $(3, 5)$ (c) $(5, 1)$ (d) $(6, -5)$
4. The reflection of the point $(4, -7)$ in the x-axis is
(a) $(-4, -7)$ (b) $(4, 7)$ (c) $(-4, 7)$ (d) $(7, 4)$

Chapter 2 · Introduction to Linear Polynomials

5. Which of the following is a linear polynomial?
(a) $x^2 + 1$ (b) $3x - 5$ (c) $1/x + 2$ (d) $\sqrt{x} + 3$
6. The zero of the polynomial $p(x) = 5x - 15$ is
(a) -3 (b) 3 (c) 5 (d) 15
7. The graph of $y = 2x + 3$ cuts the y-axis at
(a) $(3, 0)$ (b) $(0, 3)$ (c) $(0, 2)$ (d) $(2, 0)$
8. If the point $(2, k)$ lies on the line $y = 3x - 4$, then $k =$
(a) 1 (b) 2 (c) 3 (d) 10

Chapter 3 · The World of Numbers

9. Which of the following is an irrational number?
(a) $\sqrt{49}$ (b) $22/7$ (c) $\sqrt{7}$ (d) 0.25
10. The decimal expansion of $7/8$ is
(a) terminating (b) non-terminating recurring (c) non-terminating non-recurring (d) not possible
11. The value of $(2^5 \times 2^3) \div 2^4$ is
(a) 4 (b) 8 (c) 16 (d) 64
12. After rationalising the denominator, $1/(\sqrt{3} + 1)$ becomes
(a) $(\sqrt{3} - 1)/2$ (b) $(\sqrt{3} + 1)/2$ (c) $\sqrt{3} - 1$ (d) $\sqrt{3} + 1$

Chapter 4 · Exploring Algebraic Identities

13. The product $(x + 5)(x - 5)$ is equal to
(a) $x^2 + 25$ (b) $x^2 - 25$ (c) $x^2 - 10x + 25$ (d) $x^2 + 10x + 25$
14. If $a + b = 7$ and $ab = 12$, then $a^2 + b^2 =$
(a) 19 (b) 25 (c) 37 (d) 61
15. Using an identity, $103 \times 97 =$
(a) 9991 (b) 9999 (c) 10091 (d) 9909
16. The factorisation of $x^3 + 8$ is
(a) $(x + 2)(x^2 + 2x + 4)$ (b) $(x - 2)(x^2 + 2x + 4)$ (c) $(x + 2)(x^2 - 2x + 4)$ (d) $(x + 2)(x^2 - 4x + 4)$

Chapter 5 · I'm Up and Down, and Round and Round

17. The longest chord of a circle is its
(a) radius (b) diameter (c) arc (d) tangent
18. The angle subtended by a semicircle at any point on the remaining part of the circle is
(a) 60° (b) 90° (c) 120° (d) 180°

19. The sum of the opposite angles of a cyclic quadrilateral is
(a) 90° (b) 180° (c) 270° (d) 360°
20. A chord of length 16 cm is at a distance of 6 cm from the centre of a circle. The radius of the circle is
(a) 8 cm (b) 10 cm (c) 12 cm (d) 14 cm

Chapter 6 · Measuring Space: Perimeter and Area

21. The area of an equilateral triangle of side 4 cm is
(a) $4\sqrt{3} \text{ cm}^2$ (b) $8\sqrt{3} \text{ cm}^2$ (c) 16 cm^2 (d) $2\sqrt{3} \text{ cm}^2$
22. The circumference of a circle of radius 7 cm ($\pi = 22/7$) is
(a) 22 cm (b) 44 cm (c) 88 cm (d) 154 cm
23. The perimeter of a rhombus whose diagonals are 6 cm and 8 cm is
(a) 14 cm (b) 20 cm (c) 24 cm (d) 28 cm
24. The area of a triangle with sides 13 cm, 14 cm and 15 cm is
(a) 84 cm^2 (b) 91 cm^2 (c) 105 cm^2 (d) 168 cm^2

Chapter 7 · The Mathematics of Maybe: Introduction to Probability

25. The probability of a sure (certain) event is
(a) 0 (b) 1 (c) $1/2$ (d) -1
26. A die is thrown once. The probability of getting a prime number is
(a) $1/6$ (b) $1/3$ (c) $1/2$ (d) $2/3$
27. Which of the following cannot be the probability of an event?
(a) 0 (b) $3/4$ (c) $7/5$ (d) 1
28. If $P(E) = 0.35$, then $P(\text{not } E) =$
(a) 0.35 (b) 0.65 (c) 0.75 (d) 1.35

SECTION B — Fill in the Blanks

29. The point where the x-axis and the y-axis intersect is called the _____.
30. The point (0, -5) lies on the _____ axis.
31. The zero of the polynomial $4x + 12$ is _____.
32. Every point on the number line represents a unique _____ number.
33. $(a + b)^2 = a^2 + 2ab +$ _____.
34. The perpendicular drawn from the centre of a circle to a chord _____ the chord.
35. The area of a circle of radius r is _____.
36. The sum of the probabilities of all elementary events of an experiment is _____.

SECTION C — True or False

37. The point $(-2, -3)$ lies in the third quadrant.
38. The sum of two irrational numbers is always irrational.
39. $(x - y)^2 = x^2 - y^2$.
40. Angles in the same segment of a circle are equal.
41. The graph of $y = mx + c$ is a straight line.
42. The probability of an event can be 1.2.

SECTION D — Short Answer Questions

Chapter 1 · Orienting Yourself: The Use of Coordinates

43. The points A(2, 3), B(-2, 3), C(-2, -1) and D(2, -1) are joined in order. Name the figure formed and find its area.
44. Find the distance between the points P(-2, 3) and Q(4, -5).

Chapter 2 · Introduction to Linear Polynomials

45. Find the value of k if $x = 2$ is a zero of the polynomial $p(x) = 3x - k$.
46. The graph of $y = ax + 3$ passes through the point (1, 7). Find the value of a .

Chapter 3 · The World of Numbers

- 47. Express $0.777\dots$ in the form p/q .
- 48. Simplify: $81^{3/4} \times 8^{-1/3}$.
- 49. Write any two irrational numbers between 2 and 3.
- 50. Rationalise the denominator of $2/(\sqrt{5} - \sqrt{3})$.

Chapter 4 · Exploring Algebraic Identities

- 51. Expand $(2x + 3y)^2$.
- 52. Using a suitable identity, evaluate 998^2 .
- 53. Factorise: (i) $9x^2 - 16$ (ii) $x^2 + 7x + 12$
- 54. If $x + 1/x = 5$, find the value of $x^2 + 1/x^2$.

Chapter 5 · I'm Up and Down, and Round and Round

- 55. In a circle of radius 13 cm, a chord is at a distance of 5 cm from the centre. Find the length of the chord.
- 56. Two chords AB and CD of a circle with centre O are equal. If $\angle AOB = 70^\circ$, find $\angle COD$.
- 57. ABCD is a cyclic quadrilateral with $\angle A = 85^\circ$ and $\angle B = 110^\circ$. Find $\angle C$ and $\angle D$.
- 58. An arc of a circle subtends an angle of 80° at the centre. What angle does it subtend at a point on the remaining part of the circle?

Chapter 6 · Measuring Space: Perimeter and Area

- 59. Find the area of a triangle whose sides are 9 cm, 10 cm and 17 cm (use Heron's formula).
- 60. Find the area of a ring between two concentric circles of radii 14 cm and 7 cm ($\pi = 22/7$).

Chapter 7 · The Mathematics of Maybe: Introduction to Probability

- 61. A die is thrown once. Find the probability of getting (i) an even number (ii) a number greater than 4.
- 62. A bag contains 4 red, 5 blue and 3 green balls. One ball is drawn at random. Find the probability that it is (i) red (ii) not blue.

SECTION E — Long Answer Questions

Chapter 1 · Orienting Yourself: The Use of Coordinates

- 63. The vertices of a triangle are A(1, 1), B(5, 1) and C(5, 4). Show that it is a right-angled triangle and find its area and perimeter.

Chapter 2 · Introduction to Linear Polynomials

- 64. Draw the graph of $y = 2x - 1$ on graph paper. Prepare a table for $x = -1, 0, 1, 2, 3$ and state where the line cuts the axes.
- 65. A taxi charges a fixed ₹50 plus ₹12 per km. (i) Write the linear polynomial for the fare for x km. (ii) Find the fare for 8 km. (iii) For how many km is the fare ₹170?

Chapter 3 · The World of Numbers

- 66. Prove that $\sqrt{2}$ is an irrational number.

Chapter 4 · Exploring Algebraic Identities

- 67. If $a + b = 10$ and $ab = 21$, find (i) $a^3 + b^3$ (ii) $(a - b)^2$.
- 68. Factorise: (i) $x^3 - 27y^3$ (ii) $8x^3 + 12x^2 + 6x + 1$

Chapter 5 · I'm Up and Down, and Round and Round

- 69. Prove that equal chords of a circle subtend equal angles at the centre.
- 70. Prove that the opposite angles of a cyclic quadrilateral are supplementary (add up to 180°).

Chapter 6 · Measuring Space: Perimeter and Area

- 71. ABCD is a park with $\angle B = 90^\circ$, $AB = 3$ m, $BC = 4$ m, $CD = 12$ m and $DA = 13$ m. Find its area and the cost of laying turf at ₹50 per m^2 .
- 72. A circular garden of radius 14 m has a 3.5 m wide path around it. Find the area of the path and the cost of paving it at ₹10 per m^2 ($\pi = 22/7$).

Chapter 7 · The Mathematics of Maybe: Introduction to Probability

- 73.** A coin is tossed 100 times and heads appear 46 times. (i) Find the experimental probability of a head and of a tail. (ii) Find the theoretical probability of a head. (iii) Add the two experimental probabilities and comment.
- 74.** Two coins are tossed together. Write the sample space and find the probability of (i) exactly one head (ii) at least one head.

SECTION F — Case Based Question

- 75.** On a grid map of a school, the flag post is at the origin $G(0, 0)$. The library is at $L(6, 8)$, the canteen is at $C(-6, 8)$ and the playground is at $P(-6, -8)$. One unit = 1 metre.
- (i) Find the distance of the library from the flag post.
 - (ii) Find the distance between the library and the canteen.
 - (iii) Find the mid-point of the line joining the library and the canteen.
 - (iv) In which quadrant does the playground lie?

ANSWER KEY

Section A — Multiple Choice Questions

1. (b)	5. (b)	9. (c)	13. (b)	17. (b)	21. (a)	25. (b)
2. (c)	6. (b)	10. (a)	14. (b)	18. (b)	22. (b)	26. (c)
3. (c)	7. (b)	11. (c)	15. (a)	19. (b)	23. (b)	27. (c)
4. (b)	8. (b)	12. (a)	16. (c)	20. (b)	24. (a)	28. (b)

Section B — Fill in the Blanks

29. origin
30. y-axis
31. -3
32. real
33. b^2
34. bisects
35. πr^2
36. 1

Section C — True or False

37. True
38. False (e.g. $\sqrt{2} + (-\sqrt{2}) = 0$)
39. False; $(x - y)^2 = x^2 - 2xy + y^2$
40. True
41. True
42. False; probability lies between 0 and 1

Section D — Short Answer Questions

43. ABCD is a square of side 4 units; area = $4 \times 4 = 16$ sq. units.
44. $PQ = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$ units.
45. $3(2) - k = 0$, so $k = 6$.
46. $7 = a(1) + 3$, so $a = 4$.
47. Let $x = 0.777\dots$; $10x = 7.777\dots$; subtracting, $9x = 7$, so $x = 7/9$.
48. $81^{3/4} = (3^4)^{3/4} = 27$; $8^{-1/3} = 1/2$. Product = $27/2$.
49. For example $\sqrt{5} (\approx 2.236)$ and $\sqrt{7} (\approx 2.646)$. Any two valid numbers are accepted.
50. $2(\sqrt{5} + \sqrt{3})/(5 - 3) = \sqrt{5} + \sqrt{3}$.
51. $4x^2 + 12xy + 9y^2$.
52. $(1000 - 2)^2 = 1000000 - 4000 + 4 = 996004$.
53. (i) $(3x - 4)(3x + 4)$ (ii) $(x + 3)(x + 4)$
54. $x^2 + 1/x^2 = (x + 1/x)^2 - 2 = 25 - 2 = 23$.
55. Half chord = $\sqrt{(13^2 - 5^2)} = 12$ cm; chord = 24 cm.
56. Equal chords subtend equal angles at the centre, so $\angle COD = 70^\circ$.
57. $\angle C = 180^\circ - 85^\circ = 95^\circ$; $\angle D = 180^\circ - 110^\circ = 70^\circ$.
58. Angle at the circle is half the angle at the centre = 40° .
59. $s = 18$ cm; Area = $\sqrt{(18 \times 9 \times 8 \times 1)} = \sqrt{1296} = 36$ cm².
60. Area = $\pi(14^2 - 7^2) = 22/7 \times 147 = 462$ cm².
61. (i) $3/6 = 1/2$ (ii) outcomes 5, 6: $2/6 = 1/3$.
62. Total = 12. (i) $4/12 = 1/3$ (ii) $7/12$.

Section E — Long Answer Questions

63. $AB = 4$, $BC = 3$, $AC = \sqrt{4^2 + 3^2} = 5$.
 $AB^2 + BC^2 = 16 + 9 = 25 = AC^2$, so the triangle is right-angled at B.
 Area = $\frac{1}{2} \times 4 \times 3 = 6$ sq. units; Perimeter = $4 + 3 + 5 = 12$ units.
64. Table: $x = -1, 0, 1, 2, 3$ gives $y = -3, -1, 1, 3, 5$.
 Plotting and joining the points gives a straight line.

It cuts the y-axis at $(0, -1)$ and the x-axis at $(1/2, 0)$.

65. (i) $C(x) = 12x + 50$.
(ii) $C(8) = 96 + 50 = ₹146$.
(iii) $12x + 50 = 170$, so $12x = 120$ and $x = 10$ km.
66. Suppose $\sqrt{2} = p/q$, where p and q are co-prime integers and $q \neq 0$.
Then $p^2 = 2q^2$, so p^2 is even, hence p is even. Let $p = 2m$.
Then $4m^2 = 2q^2$, so $q^2 = 2m^2$, hence q is even too.
So p and q have a common factor 2, contradicting that they are co-prime. Hence $\sqrt{2}$ is irrational.
67. (i) $a^3 + b^3 = (a + b)^3 - 3ab(a + b) = 1000 - 3(21)(10) = 1000 - 630 = 370$.
(ii) $(a - b)^2 = (a + b)^2 - 4ab = 100 - 84 = 16$.
68. (i) $x^3 - 27y^3 = (x - 3y)(x^2 + 3xy + 9y^2)$.
(ii) $8x^3 + 12x^2 + 6x + 1 = (2x + 1)^3$.
69. Let AB and CD be equal chords of a circle with centre O .
In $\triangle AOB$ and $\triangle COD$: $OA = OC$ and $OB = OD$ (radii), $AB = CD$ (given).
So $\triangle AOB \cong \triangle COD$ (SSS), hence $\angle AOB = \angle COD$ (CPCT).
70. Let $ABCD$ be cyclic with centre O . Arc BCD subtends $\angle BOD$ at the centre, so $\angle BOD = 2\angle BAD$.
Arc BAD subtends the reflex angle BOD at the centre, so reflex $\angle BOD = 2\angle BCD$.
The two angles at O make 360° : $2\angle A + 2\angle C = 360^\circ$, so $\angle A + \angle C = 180^\circ$. Similarly $\angle B + \angle D = 180^\circ$.
71. $AC = \sqrt{(3^2 + 4^2)} = 5$ m; area of $\triangle ABC = \frac{1}{2} \times 3 \times 4 = 6$ m².
In $\triangle ACD$: $5^2 + 12^2 = 13^2$, so it is right-angled at C ; area = $\frac{1}{2} \times 5 \times 12 = 30$ m².
Total area = 36 m²; cost = $36 \times 50 = ₹1800$.
72. Outer radius = $14 + 3.5 = 17.5$ m.
Area = $\pi(17.5^2 - 14^2) = \frac{22}{7} \times (306.25 - 196) = \frac{22}{7} \times 110.25 = 346.5$ m².
Cost = $346.5 \times 10 = ₹3465$.
73. (i) $P(\text{head}) = 46/100 = 0.46$; $P(\text{tail}) = 54/100 = 0.54$.
(ii) Theoretical $P(\text{head}) = 1/2$.
(iii) $0.46 + 0.54 = 1$: the probabilities of all outcomes add up to 1.
74. Sample space = $\{HH, HT, TH, TT\}$.
(i) $2/4 = 1/2$.
(ii) $3/4$.

Section F — Case Based Question

75. (i) $GL = \sqrt{(36 + 64)} = 10$ m.
(ii) $LC = 6 - (-6) = 12$ m.
(iii) Mid-point = $(0, 8)$.
(iv) $(-, -)$: III quadrant.